

```
In [1]: import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: print("Seaborn version:", sns.__version__)
print("Matplotlib version:", plt.matplotlib.__version__)
```

Seaborn version: 0.13.2

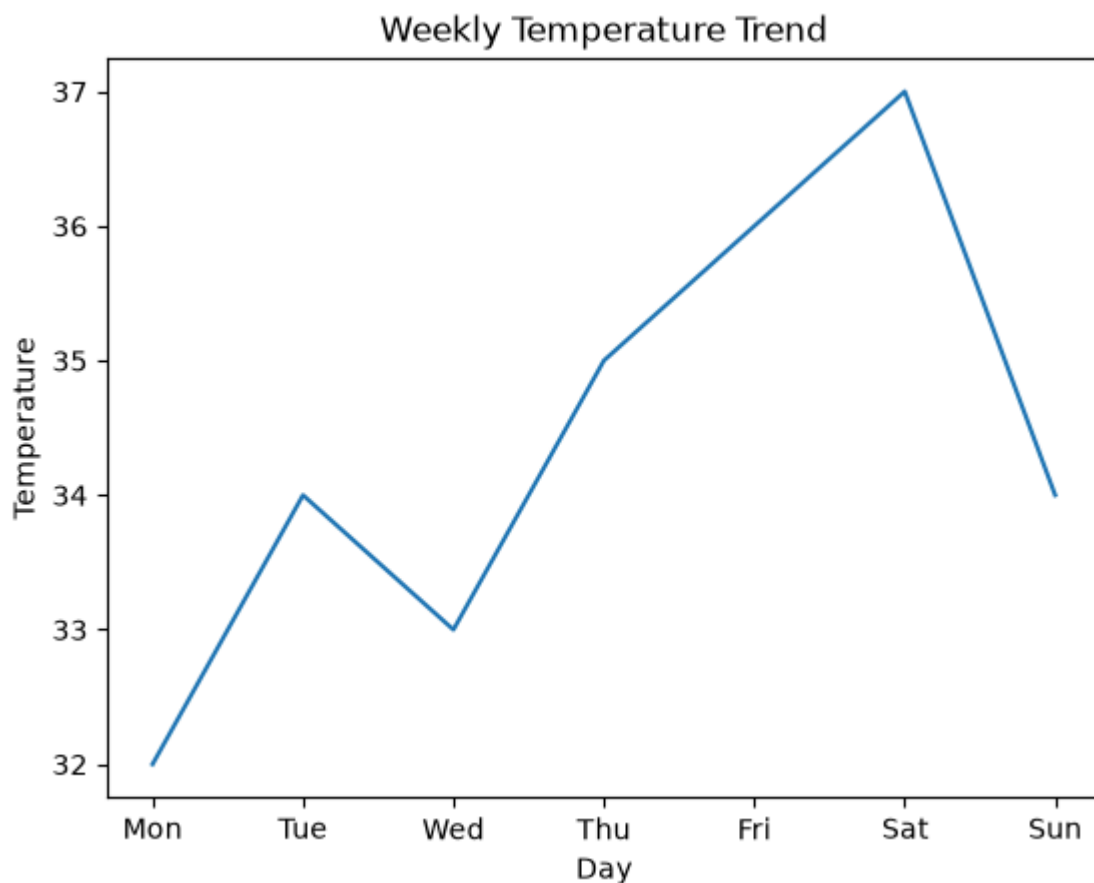
Matplotlib version: 3.11.0

```
In [3]: days = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']
temps = [32, 34, 33, 35, 36, 37, 34]
```

```
In [4]: # TASK 2 - Seaborn Line Plot

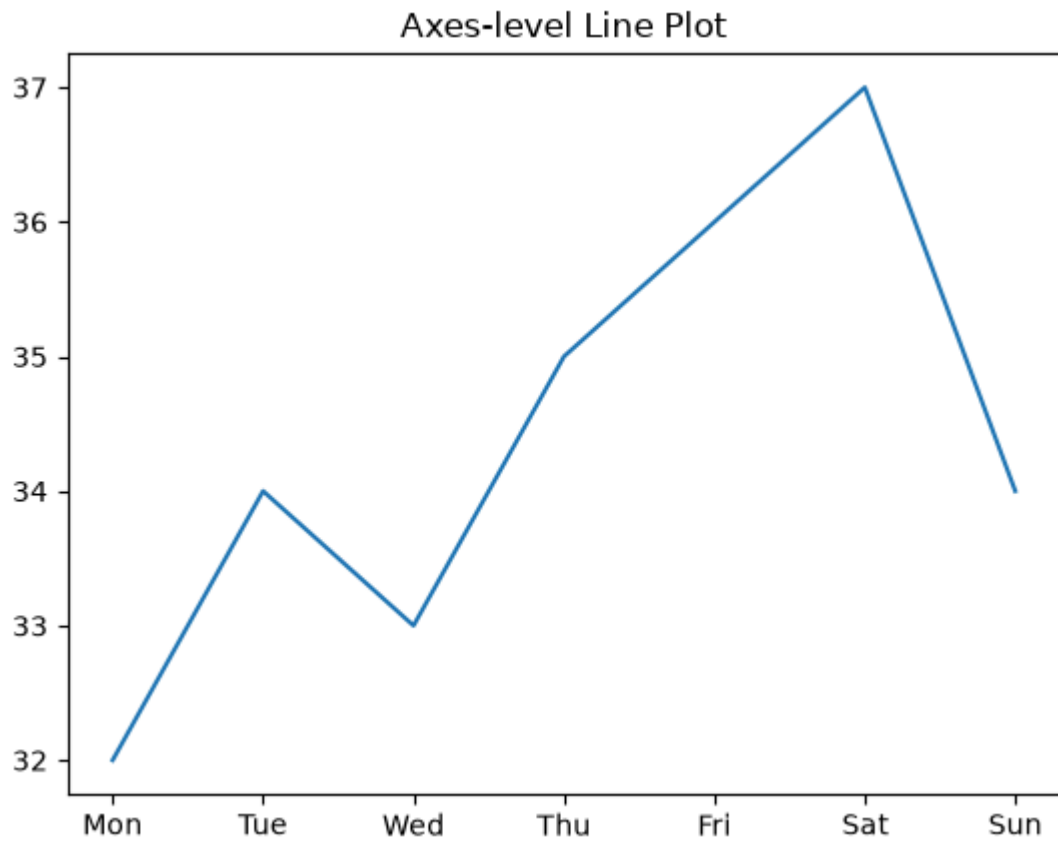
sns.lineplot(x=days, y=temps)

plt.xlabel("Day")
plt.ylabel("Temperature")
plt.title("Weekly Temperature Trend")
plt.show()
```

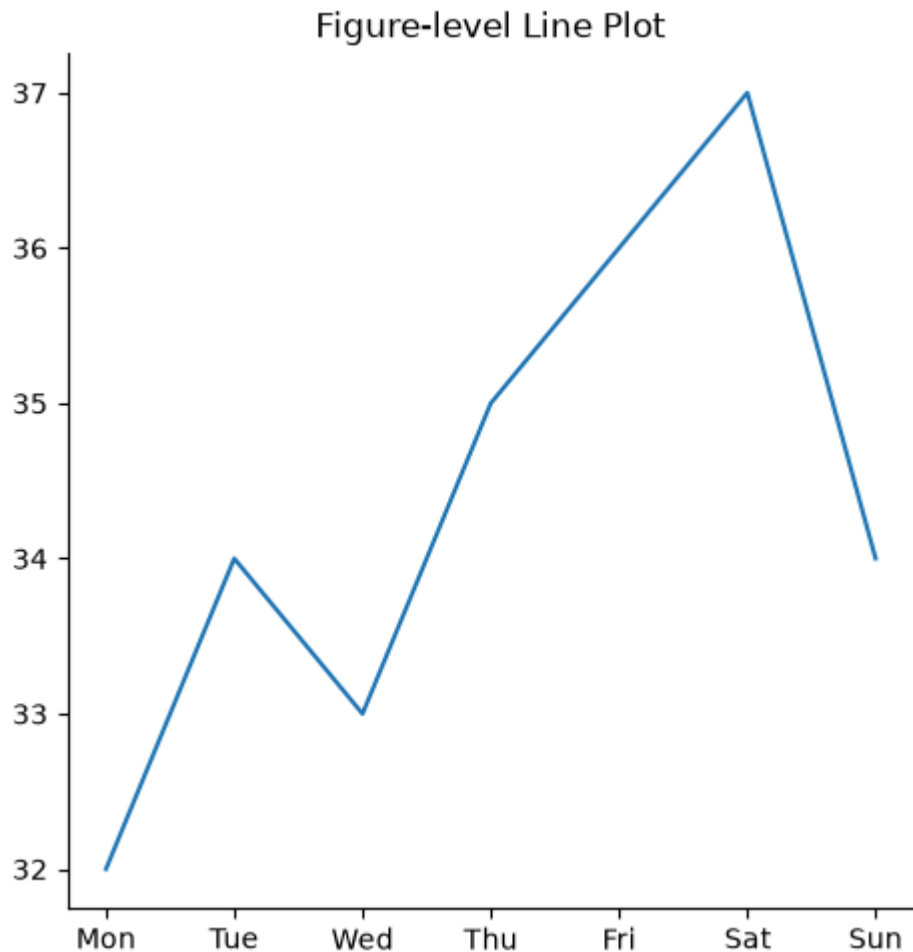


```
In [8]: # TASK 3 - Axes-Level

sns.lineplot(x=days, y=temps)
plt.title("Axes-level Line Plot")
plt.show()
```

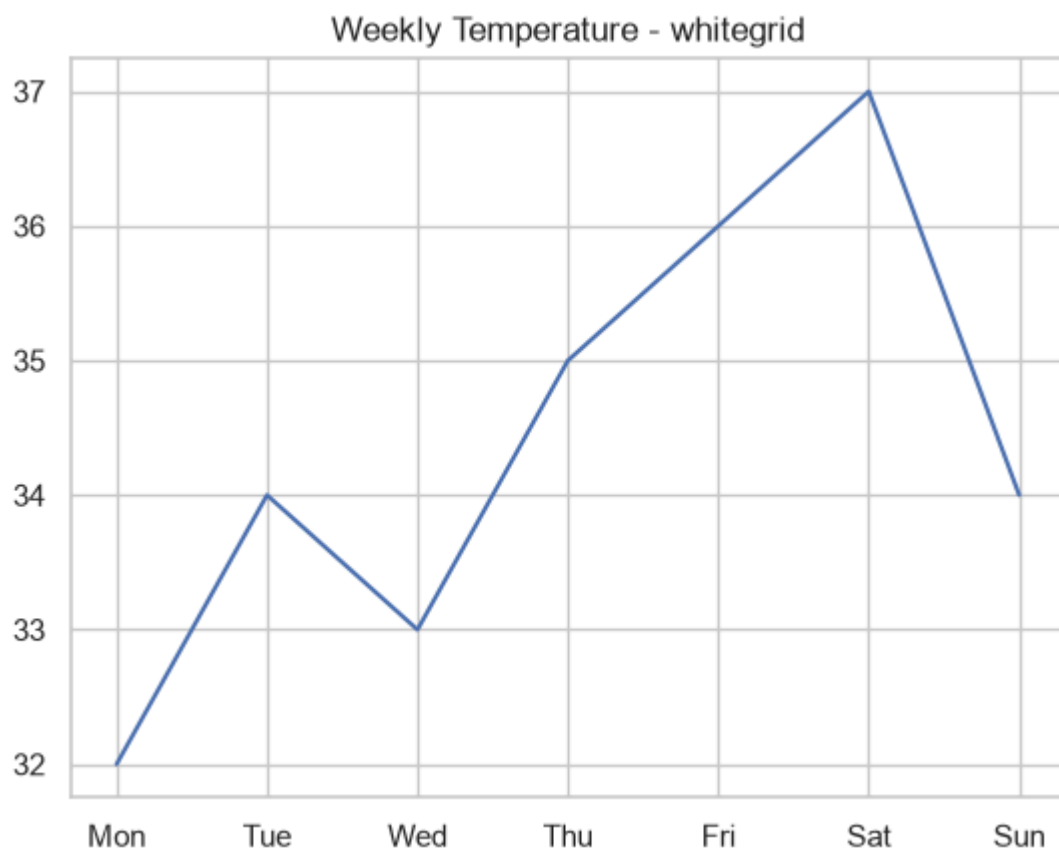
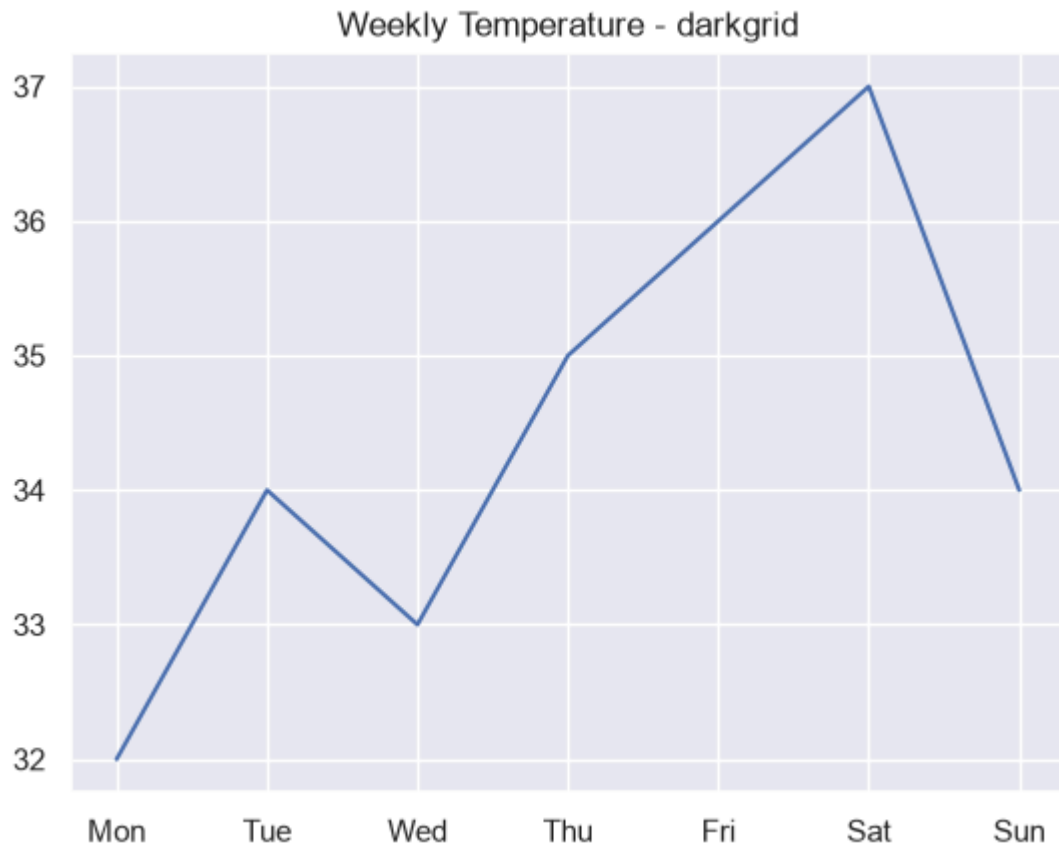


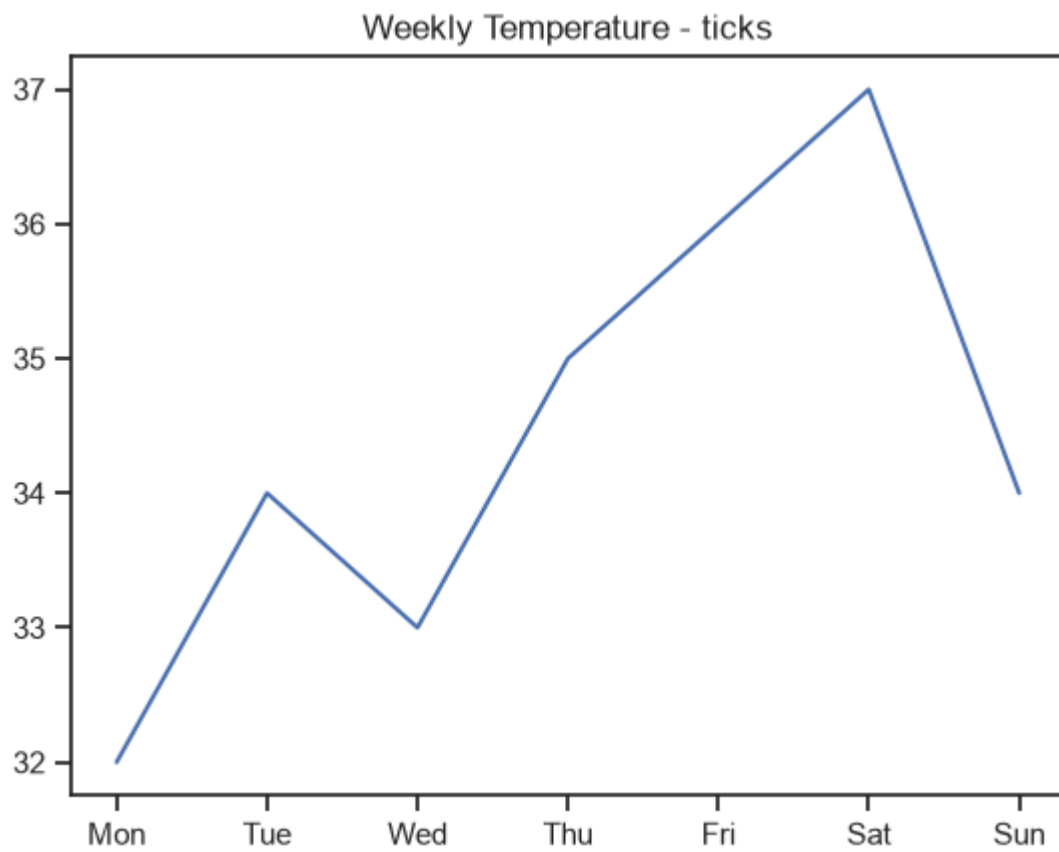
```
In [9]: # Figure-level  
  
sns.relplot(x=days, y=temps, kind="line")  
plt.title("Figure-level Line Plot")  
plt.show()
```



```
In [10]: # Analysis:  
# lineplot() works with an axes, while relplot() creates its own figure.
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```
In [11]: # TASK 4 - Seaborn Themes  
  
for theme in ["darkgrid", "whitegrid", "ticks"]:  
    sns.set_theme(style=theme)  
    sns.lineplot(x=days, y=temps)  
    plt.title("Weekly Temperature - " + theme)  
    plt.savefig(theme + ".png")  
    plt.show()  
  
# Analysis:  
# Darkgrid has a dark background with grid lines.  
# Whitegrid has a white background with grid lines.  
# Ticks has a clean background with axis ticks.
```





In []: